5-3. Regression Model Diagnostics: Residuals and Influence

Consider again the problem of fitting the model

$$y_i = a + bx_i + \varepsilon_i \quad i = 1, \ldots, n$$

Assume again a single continuous response variable y. How do we assess the quality of a given fit?

While summary statistics are helpful, they are not sufficient.

Good diagnostics are typically based on case analysis, i.e. *an examination of each observation in turn in relation to the fitting procedure*.

This leads to an examination of residuals and influence.

Residuals

The residuals should be thought of as *what is left* of the values of the response variable *after the fit has been subtracted*.

Ideally they should show no further dependence on x.

Remember: fitted equation

$$\hat{y} \quad = \quad \hat{a} + \hat{b}x$$

should capture (most of) the dependence of y upon x.

In general this should be investigated graphically by plotting residuals against the explanatory variable(s) x.

For linear models, it can also be very useful to plot residuals against fitted values $\hat{y}$ (especially when we have more than one x variable).

Example: Anscombe's Artificial Data

The R data frame `anscombe` is made available by

```
> data(anscombe)
> attach(anscombe)
```

This contains 4 artificial datasets created by Frank Anscombe, each of 11 observations of a continuous response variable y and a continuous explanatory variable x. A full discussion is given in *Applied Linear Regression*, by S. Weisberg (1985, p107-108)

We fit a least squares linear model to each data set.

Take a look at the data:

```
> anscombe
```

```
   x1 x2 x3 x4    y1   y2    y3    y4
1  10 10 10  8  8.04 9.14  7.46  6.58
2   8  8  8  8  6.95 8.14  6.77  5.76
3  13 13 13  8  7.58 8.74 12.74  7.71
4   9  9  9  8  8.81 8.77  7.11  8.84
5  11 11 11  8  8.33 9.26  7.81  8.47
6  14 14 14  8  9.96 8.10  8.84  7.04
7   6  6  6  8  7.24 6.13  6.08  5.25
8   4  4  4 19  4.26 3.10  5.39 12.50
9  12 12 12  8 10.84 9.13  8.15  5.56
10  7  7  7  8  4.82 7.26  6.42  7.91
11  5  5  5  8  5.68 4.74  5.73  6.89
```

Model 1

```
> model1 = lm(y1 ~ x1)
> summary(model1)

Call:
lm(formula = y1 ~ x1)

Residuals:
     Min       1Q   Median       3Q      Max
-1.92127 -0.45577 -0.04136  0.70941  1.83882

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.0001     1.1247   2.667  0.02573 *
x1            0.5001     0.1179   4.241  0.00217 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.237 on 9 degrees of freedom
Multiple R-squared:  0.6665,Adjusted R-squared:  0.6295
F-statistic: 17.99 on 1 and 9 DF,  p-value: 0.00217
```

Model 2

```
> model2 = lm(y2 ~ x2)
> summary(model2)

Call:
lm(formula = y2 ~ x2)

Residuals:
    Min      1Q  Median      3Q     Max
-1.9009 -0.7609  0.1291  0.9491  1.2691

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    3.001      1.125   2.667  0.02576 *
x2             0.500      0.118   4.239  0.00218 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.237 on 9 degrees of freedom
Multiple R-squared:  0.6662,Adjusted R-squared:  0.6292
F-statistic: 17.97 on 1 and 9 DF,  p-value: 0.002179
```

Model 3

```
> model3 = lm(y3 ~ x3)
> summary(model3)

Call:
lm(formula = y3 ~ x3)

Residuals:
    Min      1Q  Median      3Q     Max
-1.1586 -0.6146 -0.2303  0.1540  3.2411

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.0025     1.1245   2.670  0.02562 *
x3            0.4997     0.1179   4.239  0.00218 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.236 on 9 degrees of freedom
Multiple R-squared:  0.6663,Adjusted R-squared:  0.6292
F-statistic: 17.97 on 1 and 9 DF,  p-value: 0.002176
```

Model 4

```
> model4 = lm(y4 ~ x4)
> summary(model4)

Call:
lm(formula = y4 ~ x4)

Residuals:
   Min     1Q Median     3Q    Max
-1.751 -0.831  0.000  0.809  1.839

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.0017     1.1239   2.671  0.02559 *
x4            0.4999     0.1178   4.243  0.00216 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.236 on 9 degrees of freedom
Multiple R-squared:  0.6667,Adjusted R-squared:  0.6297
F-statistic:    18 on 1 and 9 DF,  p-value: 0.002165
```

In each case, the fitted line is $\hat{y} = 3.00 + 0.50x$. We also find that the standard errors of $\hat{a}, \hat{b}$, the residual standard error, and the coefficient of determination R^2 are each *almost identical* across the 4 datasets.

Judging by everything we've seen so far, it appears that all four data sets are equally well described by the same linear regression model.

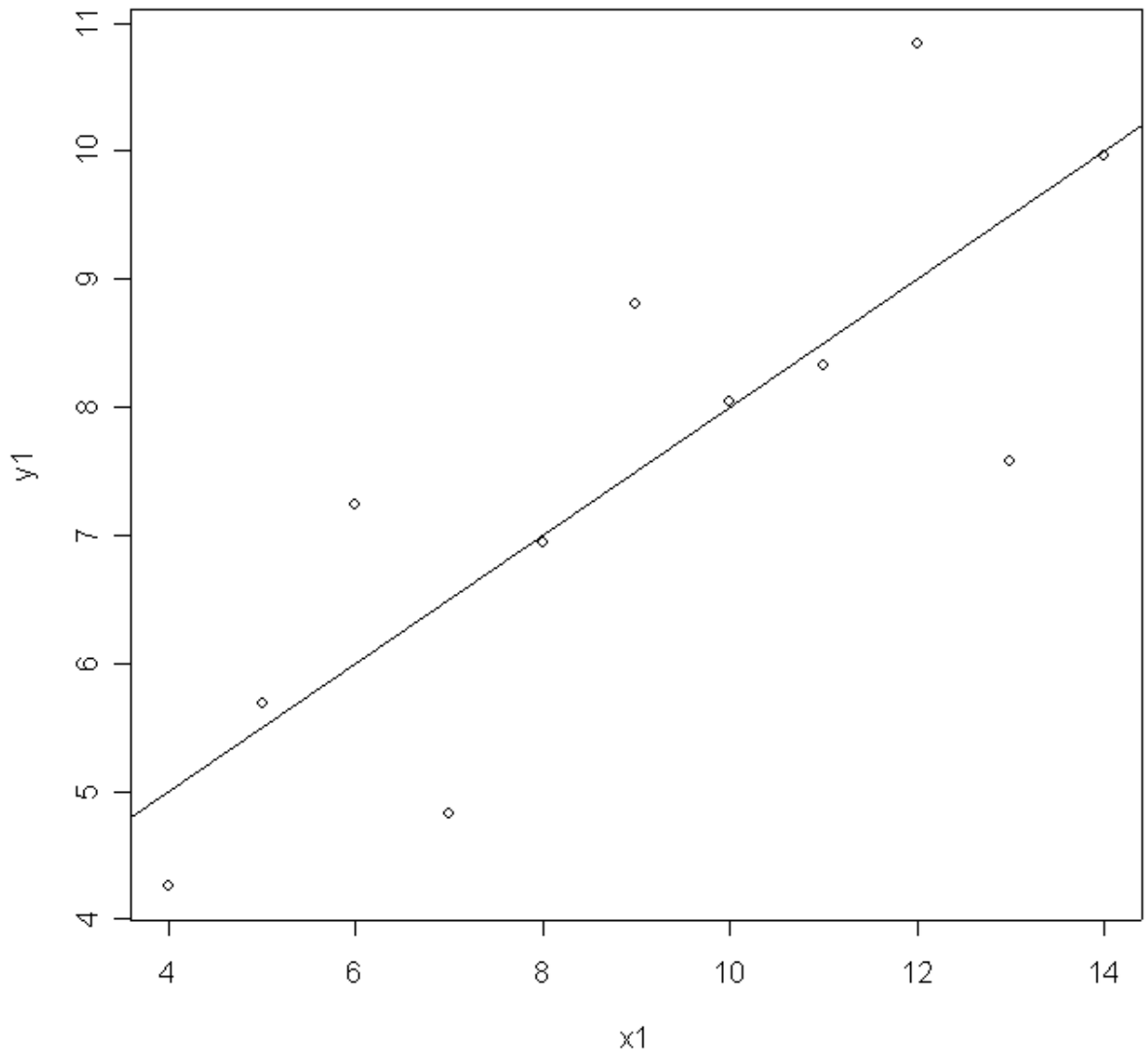
But now we'll look at some graphs, and find that very *different* judgements should be made about the appropriateness of the fitted model to each of the 4 cases. We look at scatterplots of the data, and also plots of residuals.

Recall model assumption: $\epsilon_i \sim N(0,\ \sigma^2)$, independent

So we want plotted residuals e_i to be

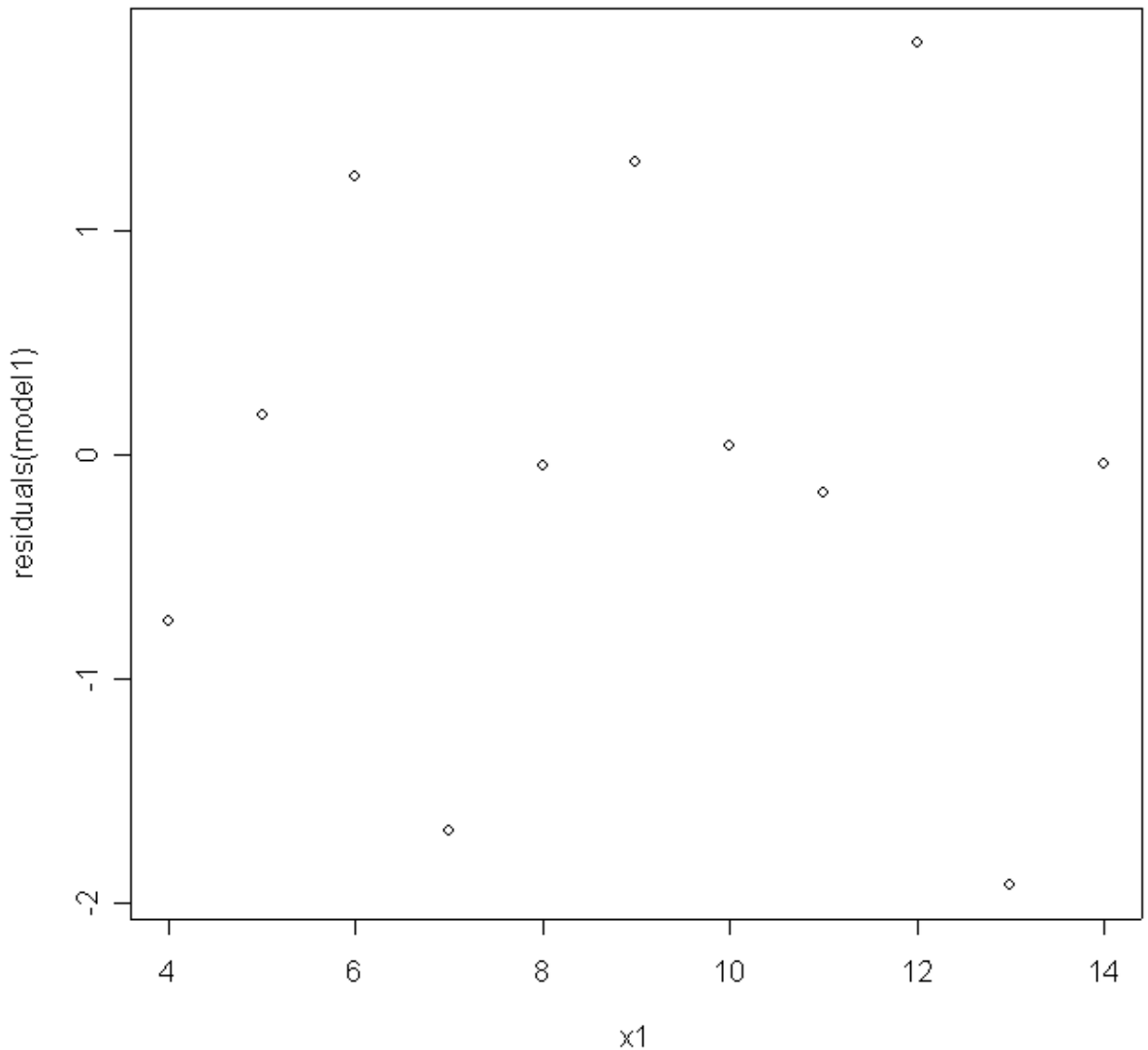
- centred about 0
- with constant variance (so the amount of vertical spread in the residuals does not change as we move along the horizontal axis)

Model 1: $y1$ vs $x1$ and fitted line



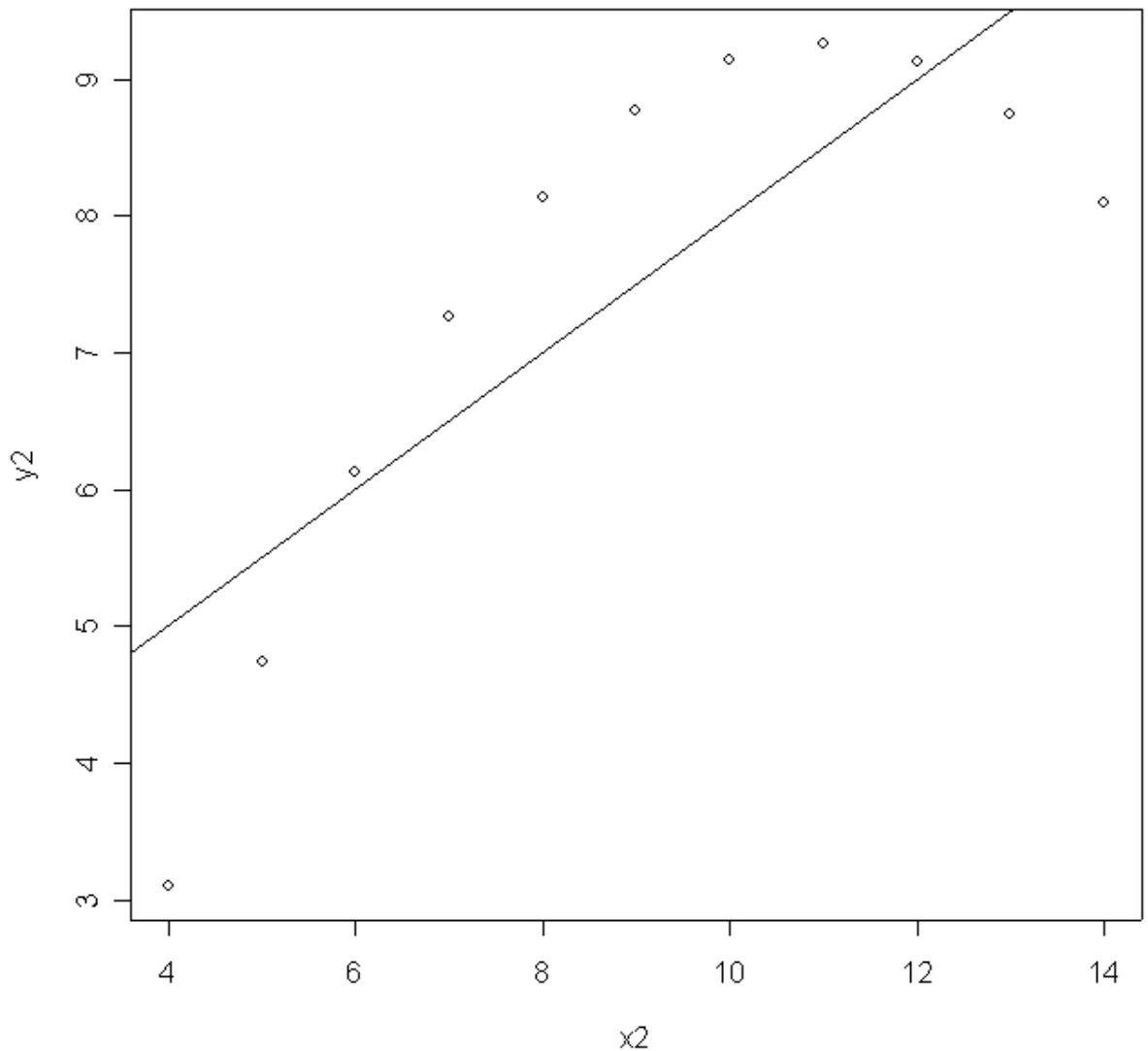
Reasonably good fit.

Model 1: residuals



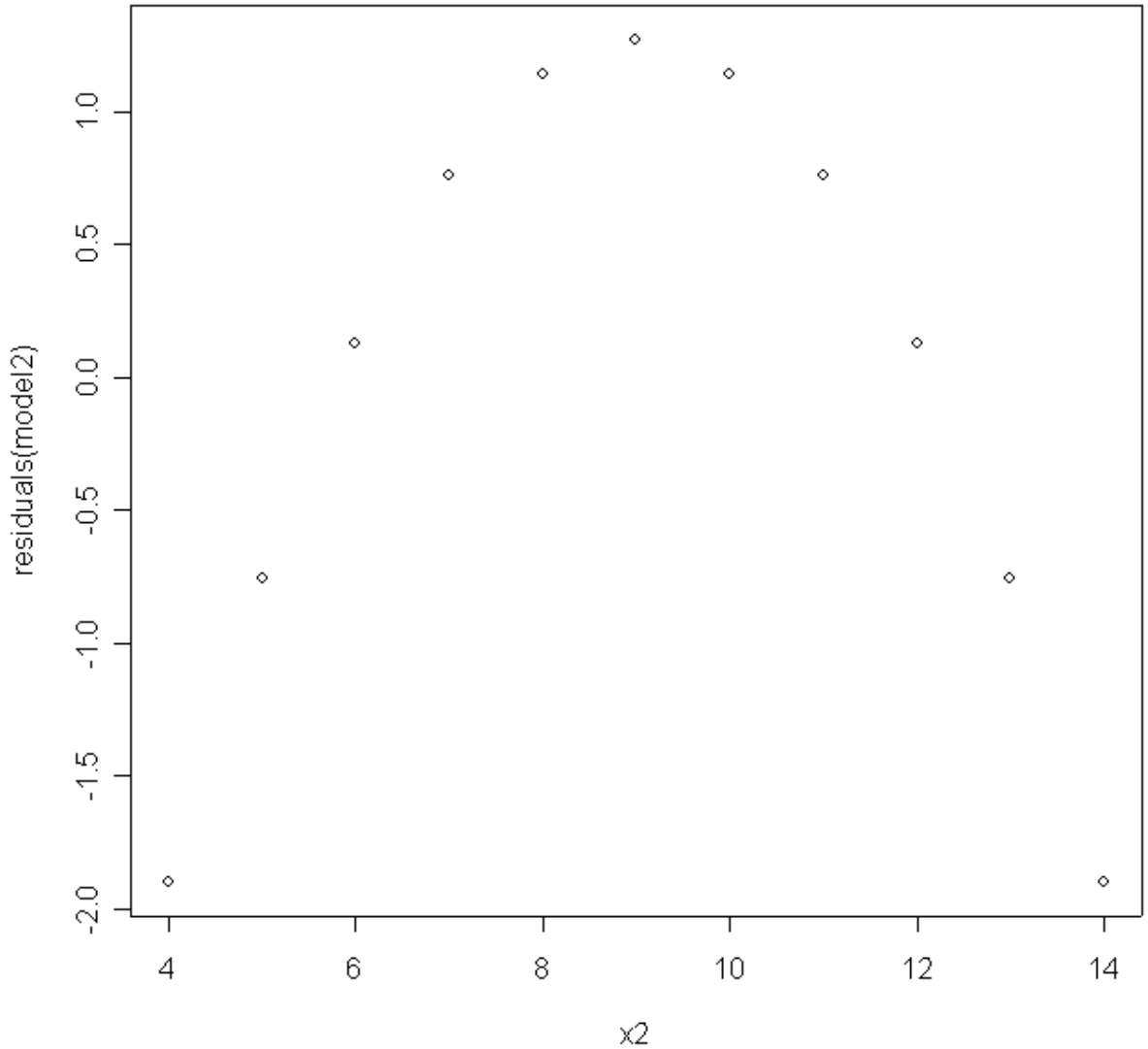
No particular pattern; model is fine.

Model 2: $y2$ vs $x2$ and fitted line



Non-linear relationship; need a non-linear model.

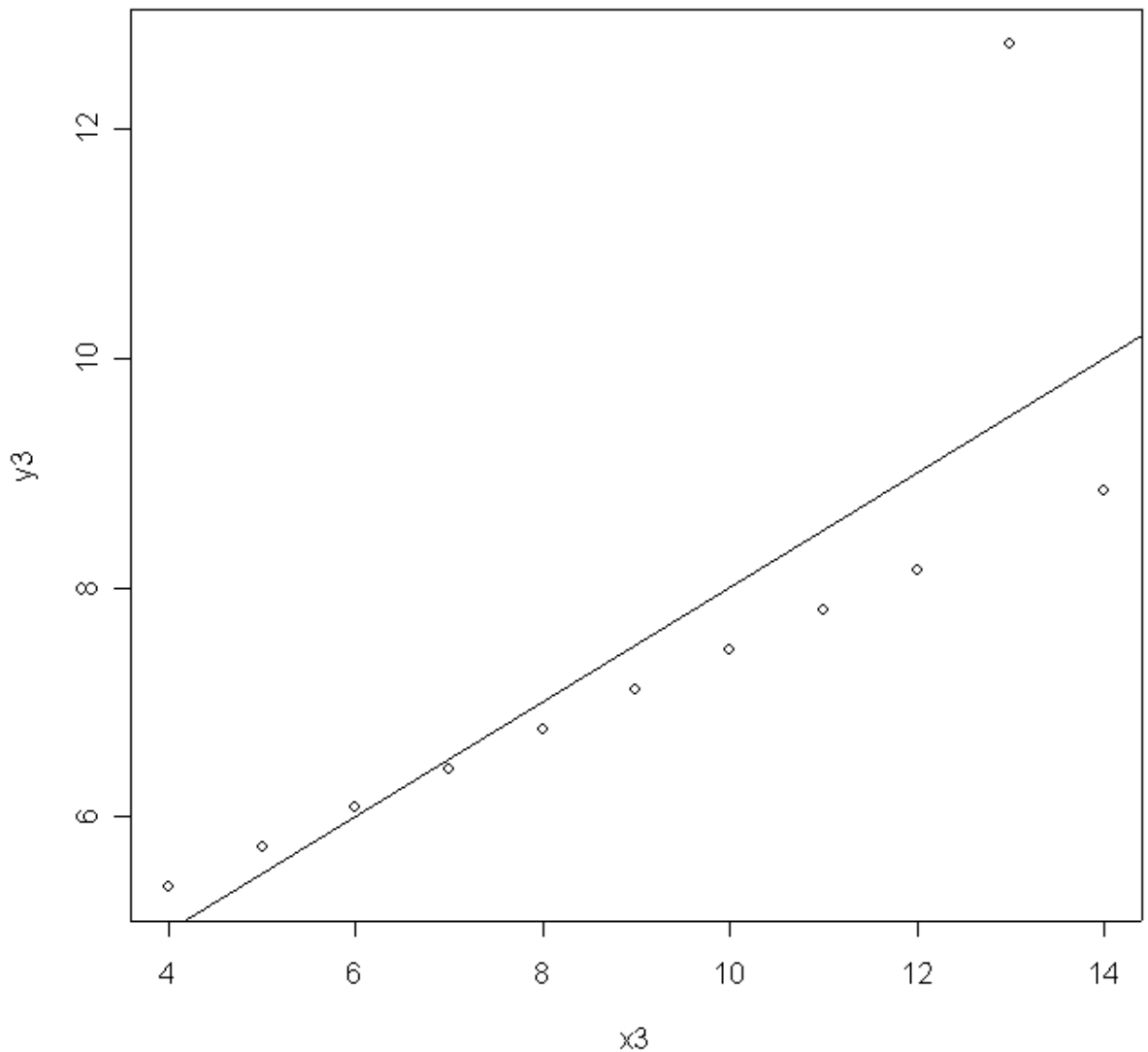
Model 2: residuals



Residuals depend systematically on x (possibly quadratic relationship).

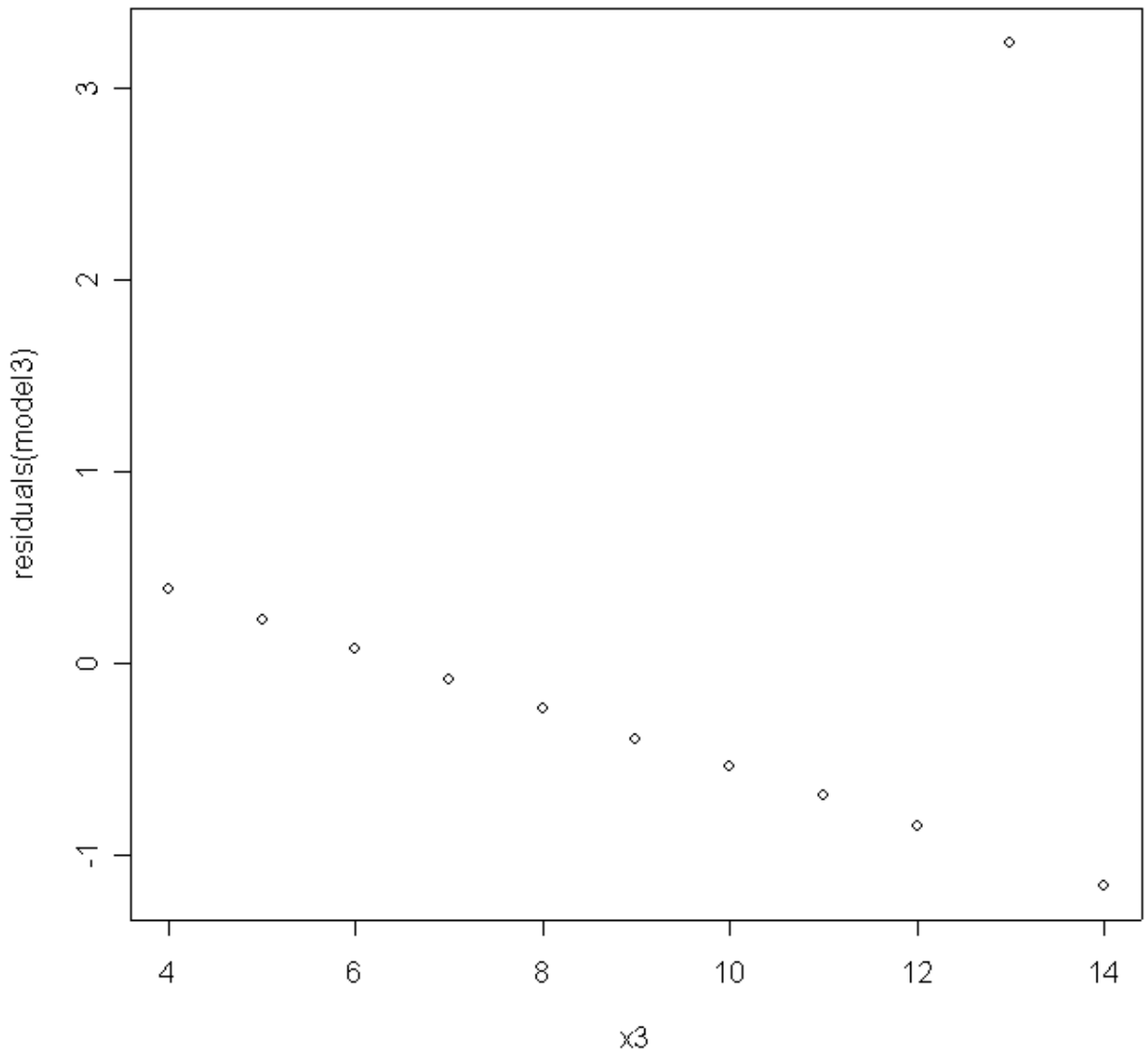
Fitted line $\hat{y} = \hat{a} + \hat{b}x$ does not capture all dependence of y on x

Model 3: $y3$ vs $x3$ and fitted line



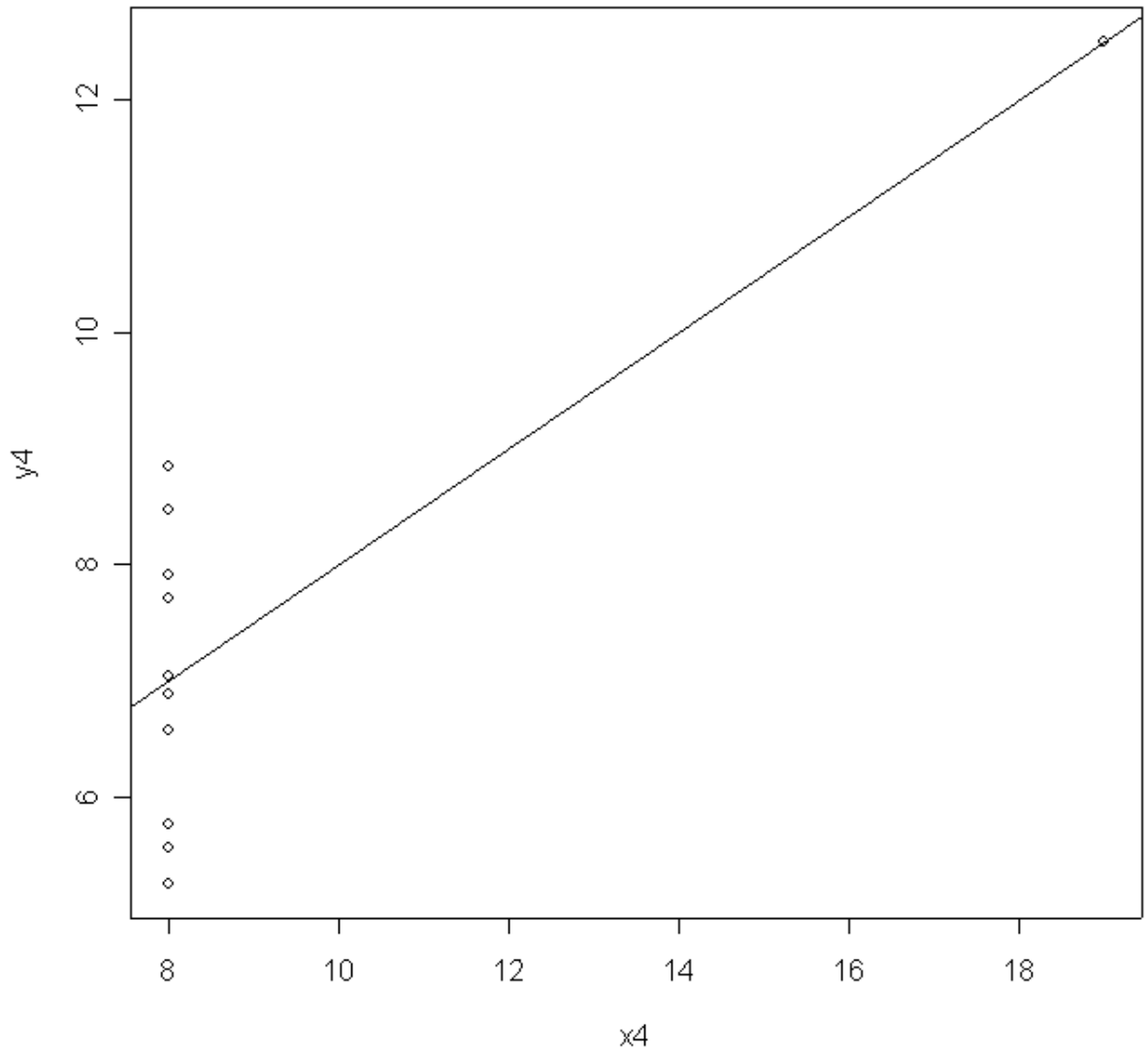
Perfect linear relationship except for one *outlier*. Should treat this point separately from the rest.

Model 3: residuals



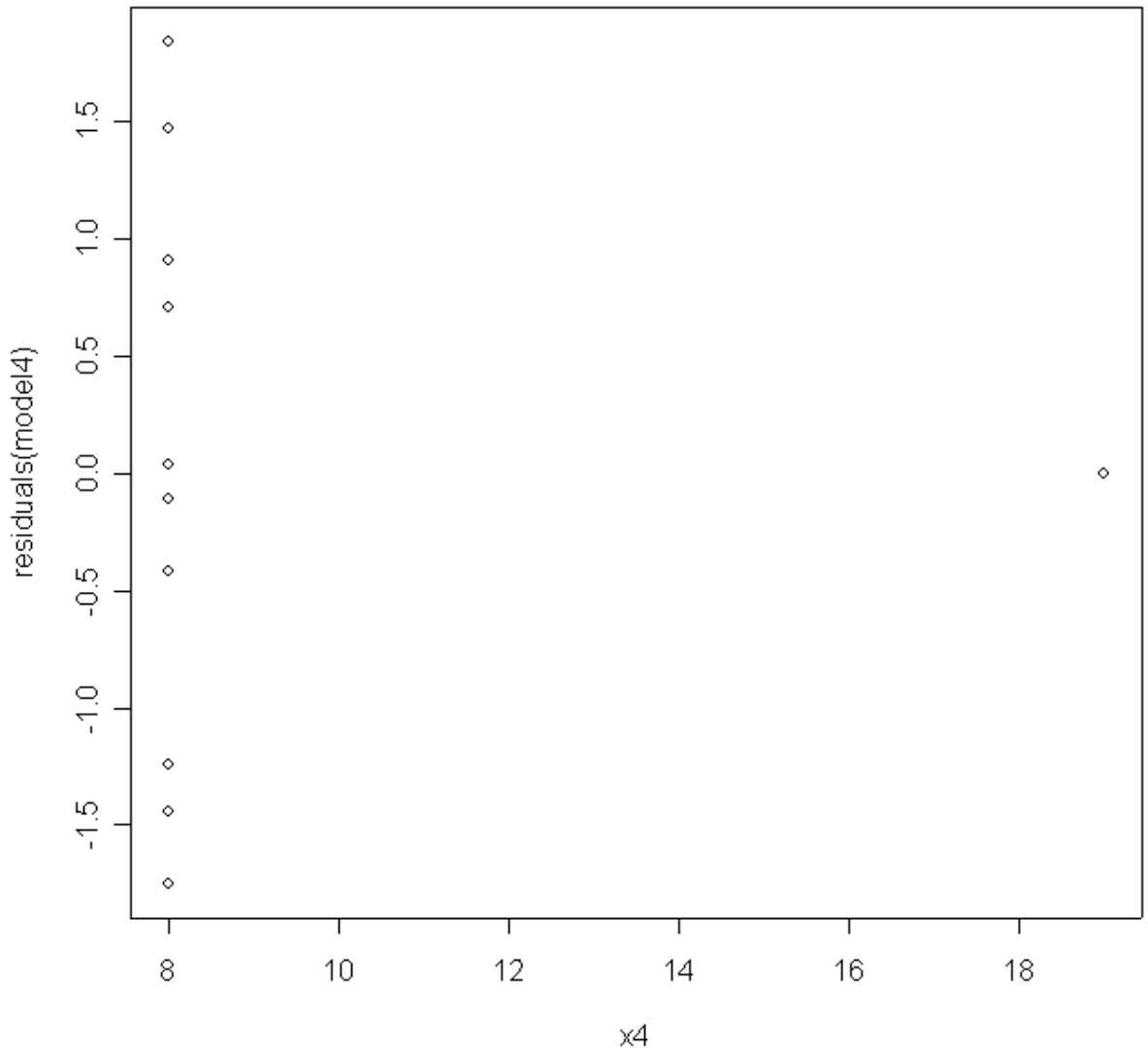
Trend in residuals (decrease as x increases), and *outlier*.

Model 4: $y4$ vs $x4$ and fitted line



Only two x values. Can't tell whether the true relationship is linear or not.

Model 4: residuals

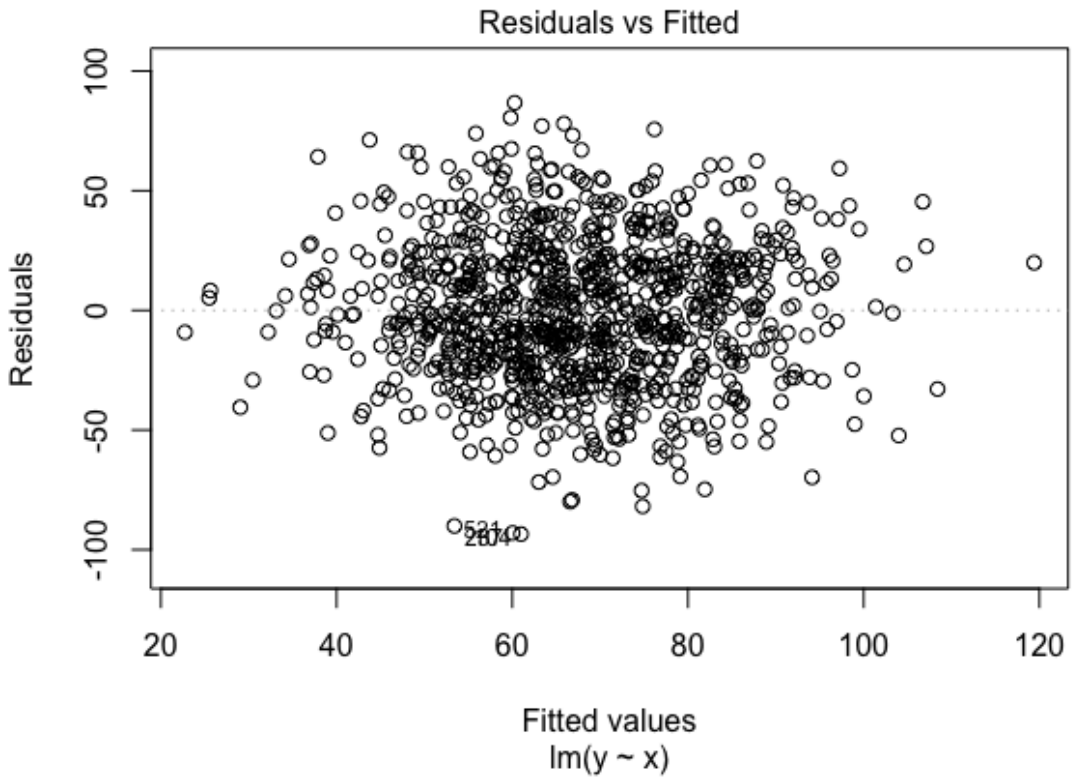


In general, residuals provide information about:

- whether the *best relation* has been fitted
- the relative merits of *different fits*
- mild, but non-random, *departures from the hypothesised fit*
- the magnitude of the residual variation
- the identification of *outliers*
- possible further dependence on x, other than through location, of the conditional distribution of y given x
  – in particular *heterogeneity of spread* of the residuals.

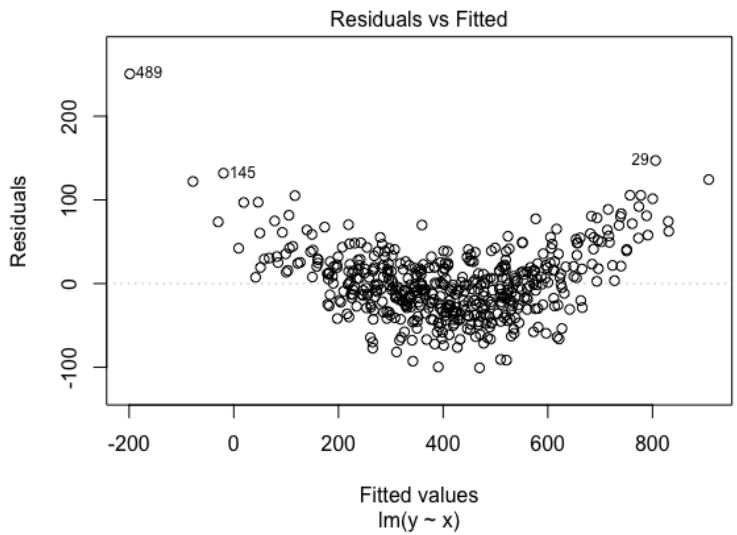
Plotting residuals against fitted values can show up problems with the model.

A satisfactory residuals plot:

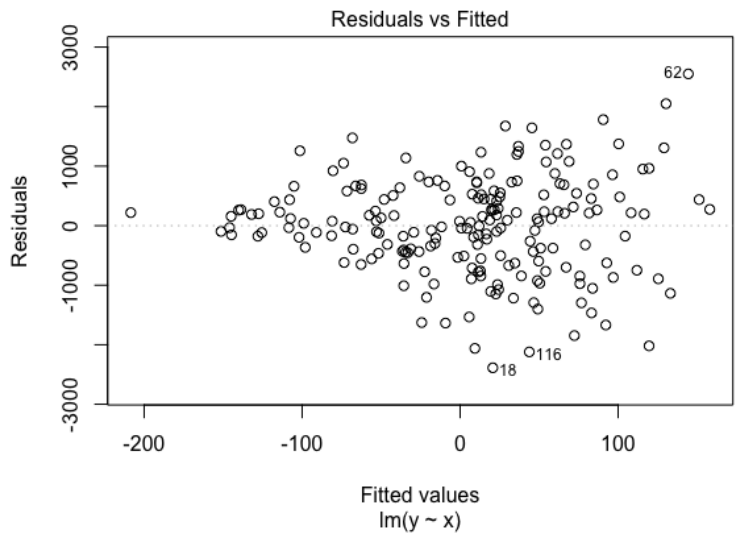


No pattern. Random scatter, residuals centred about 0.

Problematic residuals plots:



Non-linear relationship.



Heteroscedasticity.

Heteroscedasticity: vertical spread of residuals changes as fitted values increase.

Standardised residuals

$$r_i = \frac{e_i}{\hat{\sigma}\sqrt{1 - h_{ii}}}$$

where h_{ii} is the i-th diagonal element of the "hat" matrix (see later).

In simple linear regression:

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$$

Under model assumptions:

- $E(r_i) = 0$ (as for e_i)
- But all have same variance, $var(r_i) = 1$
  Note that $var(e_i) = \sigma^2 (1 - h_{ii})$

Influence

Influence measures the extent to which a fit is affected by individual observations.

A possible formal definition is the following: the influence of any observation is a measure of the difference between (i) the fit and (ii) the fit which would be obtained if that observation were omitted.

Observations with large influences require more careful checking.

Especially for linear models, influence is often measured by Cook's distance.

Cook's Distance

$$D_i = \frac{\sum_{j=1}^{n} \left(\hat{y}_j^{(i)} - \hat{y}_j\right)^2}{p\,\widehat{\sigma^2}}$$

- $\hat{y}_j$ is the fitted value of observation j where the fit is obtained using all the data
- $\hat{y}_j^{(i)}$ is the fitted value of observation j where the fit is obtained using all the data except observation i
- p is the number of regression parameters in the model (so $p = 2$ for simple linear regression)

This will require running $(n + 1)$ models to obtain all D_i values. Instead we can use the equivalent expression:

$$D_i = \frac{r_i^2}{p}\frac{h_{ii}}{1 - h_{ii}} = \frac{e_i^2}{p\,\widehat{\sigma^2}}\,\frac{h_{ii}}{(1 - h_{ii})^2}$$

where e_i and r_i are the usual and standardised residuals respectively, and h_{ii} is the ith diagonal element of the hat matrix H (defined later).

As a rule of thumb, observations for which $\boxed{D_i > 1}$ make a noticeable difference to the parameter estimates, and should be examined carefully for the appropriateness of their use in fitting the model.

An observation with a large residual *usually* has a large influence, but not always.

However, an observation with an unusual value of its explanatory variable(s) x can pull a fit towards it and have a large influence though a small residual.

We need to look at *both* residuals and influence to check the regression.

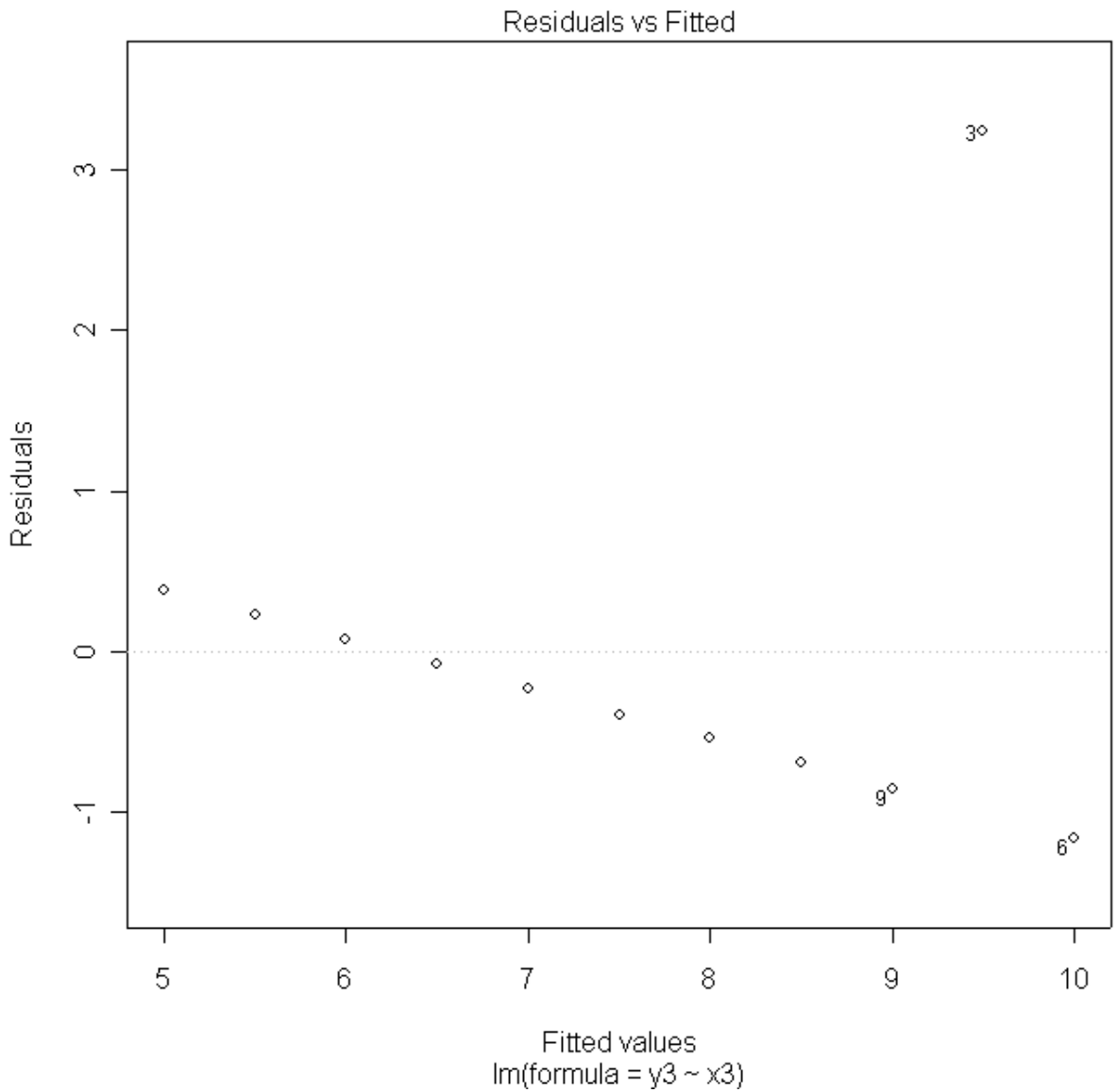
Example: Anscombe's third data set.

```
> plot(model3, which = 1:4)
```

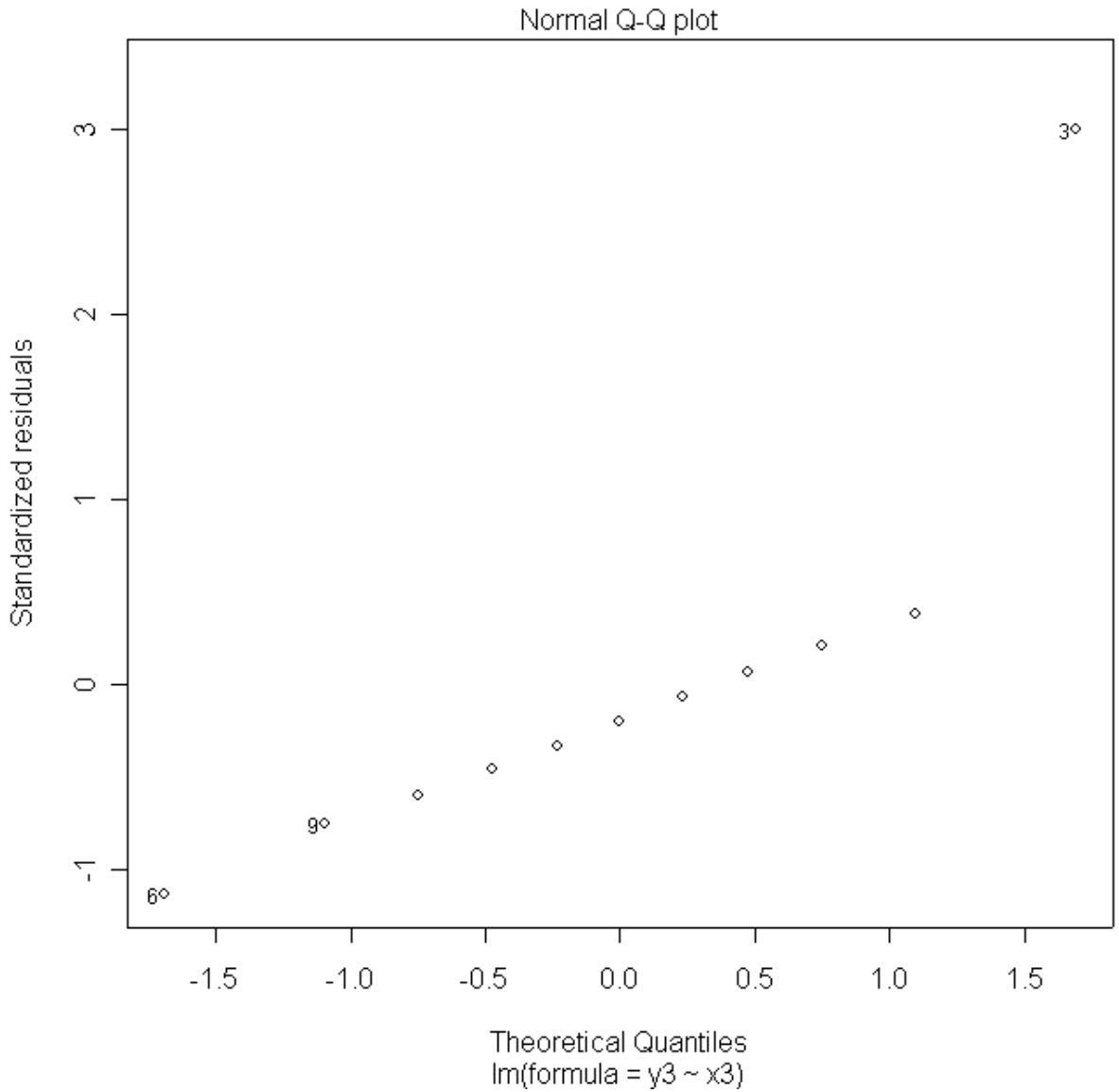
Take a look at R help using `?plot.lm`

Produces the following 4 graphs:

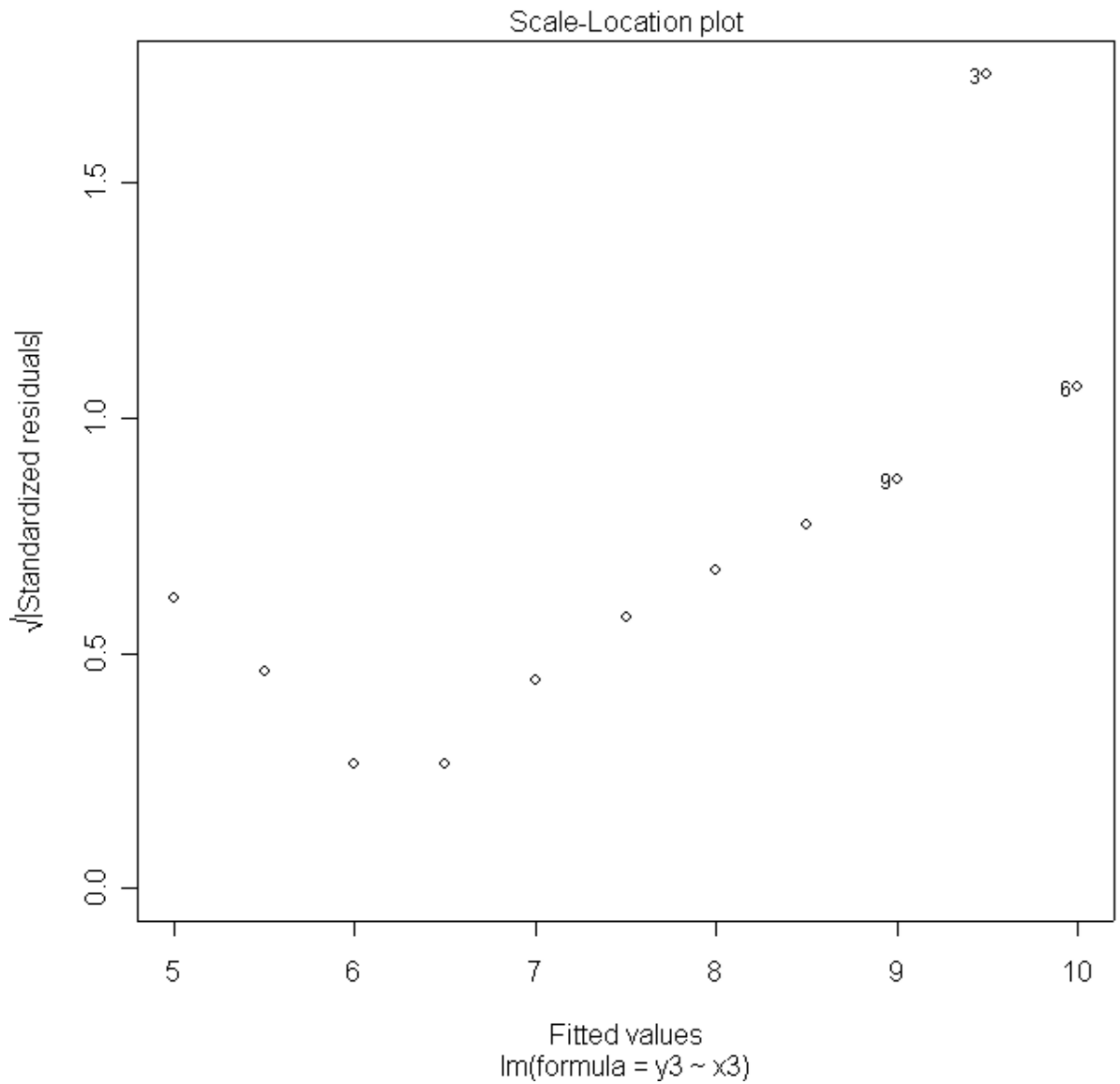
Model 3: residuals



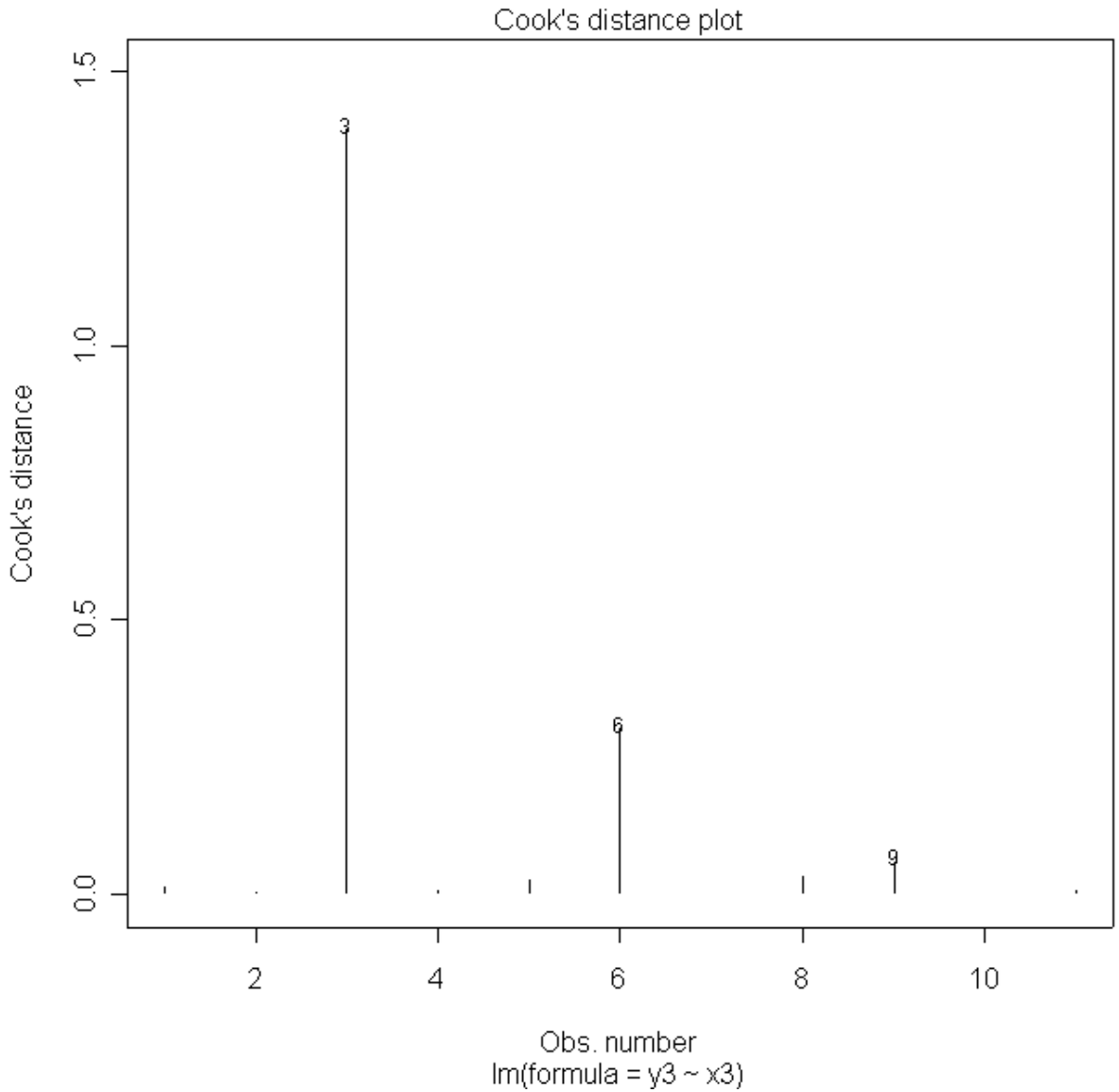
Model 3: normal Q-Q plot



Model 3: standardised residuals



Model 3: Cook's distance plot



The last graph produced by the plot function shows that observation number 3 has an unusually large value of Cook's distance

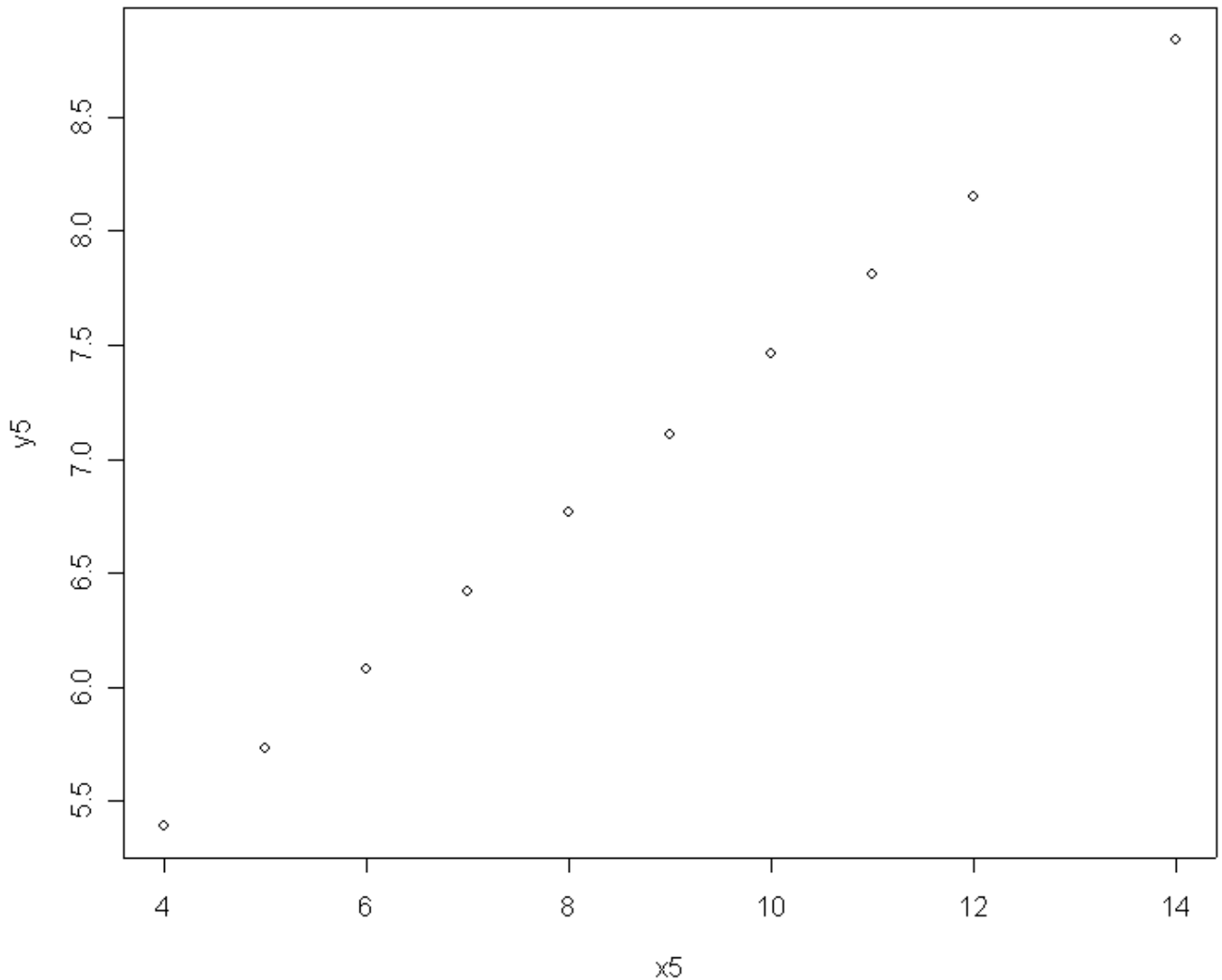
$$D_3 = 1.39$$

We now refit the data omitting this observation:

```
> x5=x3[-3]
> y5=y3[-3]
> model5=lm(y5~x5)
```

Note: we investigate the effect of omitting this data point; we do not simply throw it away and forget about it.

Model 5: $y5$ vs $x5$



Model 5

```
> summary(model5)

Call:
lm(formula = y5 ~ x5)

Residuals:
       Min          1Q      Median          3Q         Max
-0.0041558 -0.0022240  0.0000649  0.0018182  0.0050649

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 4.0056494  0.0029242    1370   <2e-16 ***
x5          0.3453896  0.0003206    1077   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.003082 on 8 degrees of freedom
Multiple R-squared:      1,Adjusted R-squared:      1
F-statistic: 1.161e+06 on 1 and 8 DF,  p-value: < 2.2e-16
```

For the full data set, the fitted model 3 was

$$\hat{y} = 3.0025 + 0.4997x$$

With observation 3 omitted, the new fitted model 5 is

$$\hat{y} = 4.0056 + 0.3454x$$

Note substantial changes in estimated intercept and estimated slope, confirming that observation 3 is *influential*.

Further regression example: Brain and Body Weights

Brain weights (g) and body weights (kg) for 62 species of mammal ranging in size from the lesser short-tailed shrew (0.005 body weight) to the African elephant (6654 kg body weight).

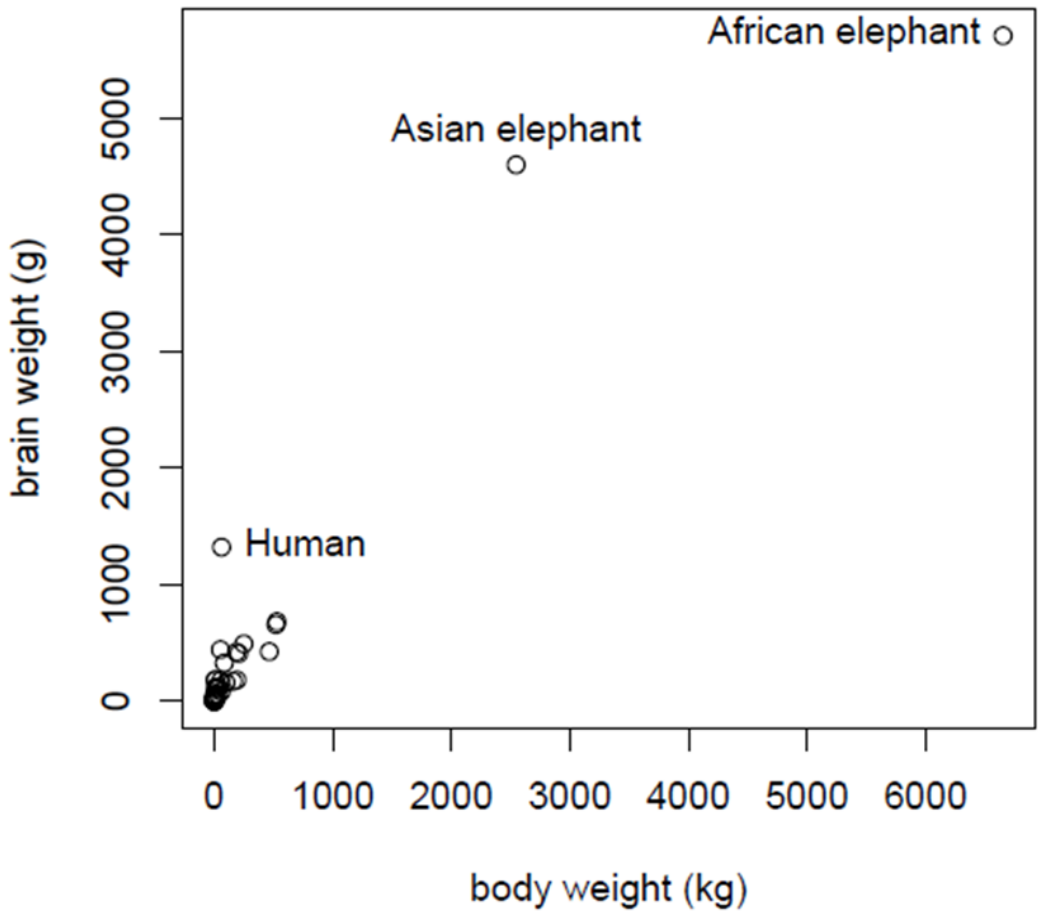
The data are taken from a study by Allison and Cicchetti (1976), and are available in R by:

```
> library(MASS)
> data(mammals)
```

Here brain weight is treated as the **response variable** and body weight as the **explanatory variable**.

The following `R` code produces the plot of brain weight against body weight. Note that the `identify` command requires appropriate user interaction, to identify interesting points in the plot.

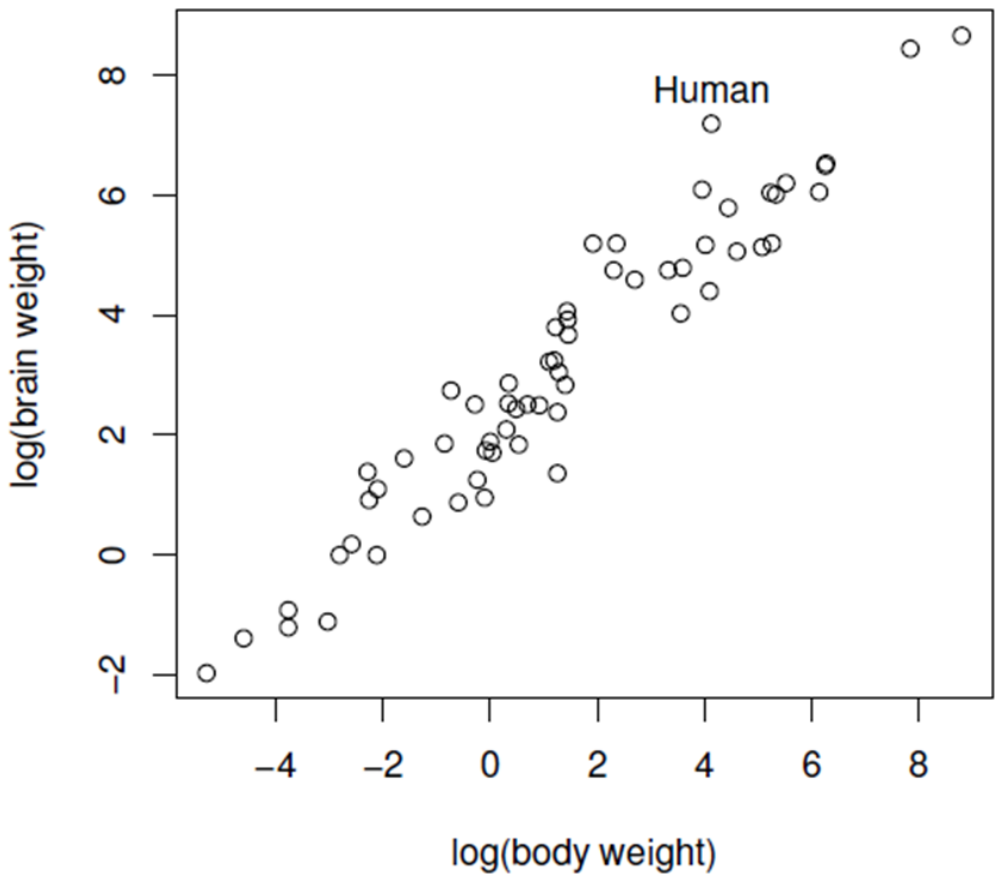
```
> plot(brain ~ body, data=mammals, xlab = "body weight (kg)",
 ylab = "brain weight (g)")
> attach(mammals)
> identify(brain ~ body, labels = row.names(mammals))
```



The plot suggests transformation of the data will be helpful. Severe positive **skewness** of the distributions of both variables suggests **log transformation** for both.

Plot transformed data; remember that `identify` command requires user interaction.

```
> plot(log(brain) ~ log(body), data=mammals,
xlab = "log(body weight)", ylab = "log(brain weight)")
> identify(log(brain) ~ log(body), labels = row.names(mammals))
```



Fit the linear regression model to transformed data:

$$\log(\text{brainweight}) = a + b \times \log(\text{bodyweight}) + \varepsilon$$

In R:

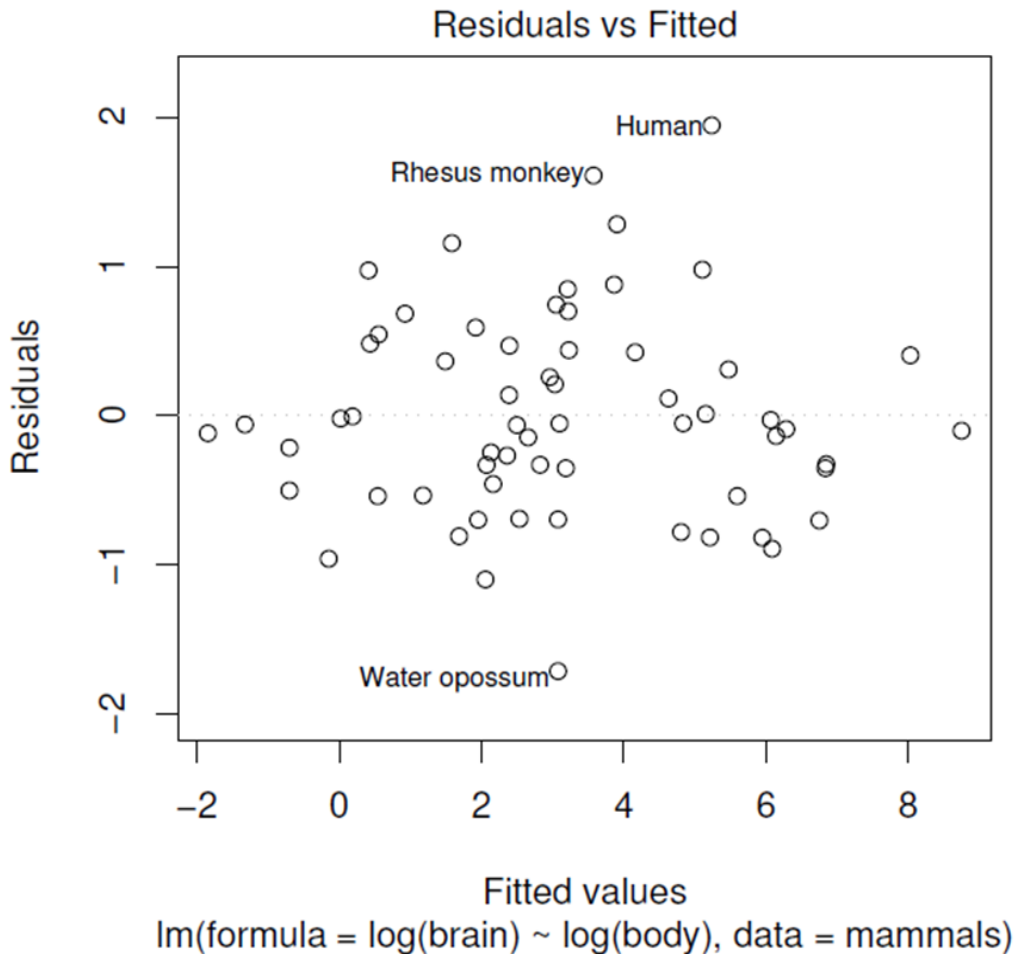
```
> mammals.model.1 = lm(log(brain) ~ log(body))
> summary(mammals.model.1)
> plot(mammals.model.1)
```

The fitted model is

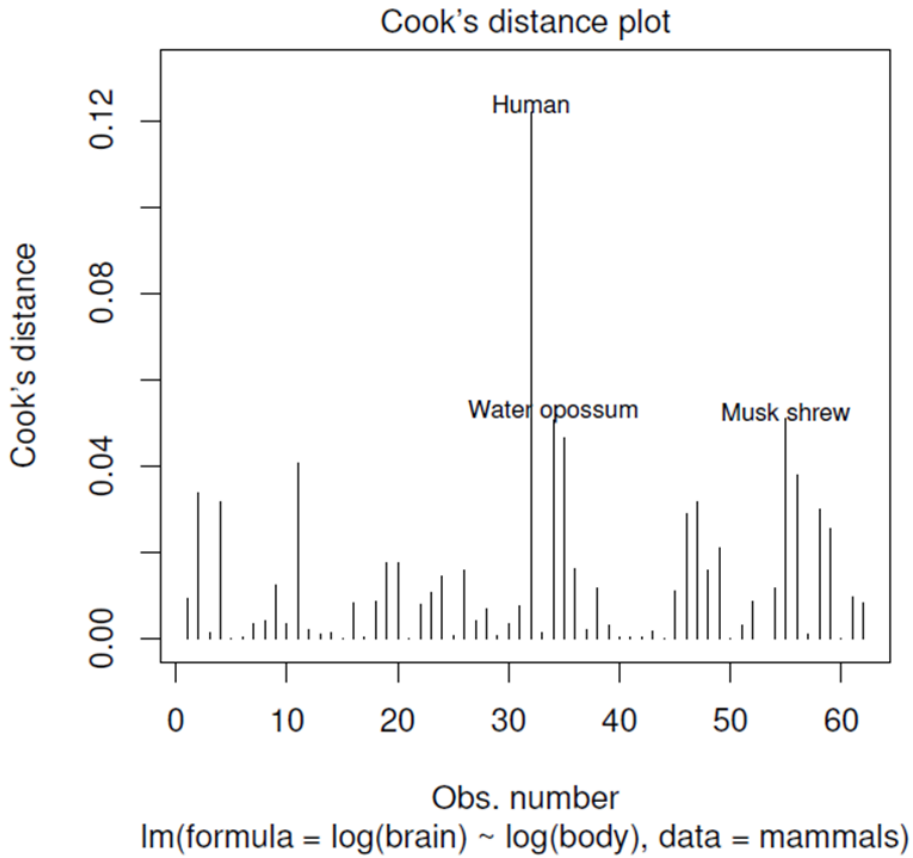
$$\log(\text{brainweight}) = 2.13 + 0.752 \log(\text{bodyweight})$$

In terms of the untransformed variables,

$$\text{brainweight} = 8.46 \times \text{bodyweight}^{0.752}$$



Model fit seems satisfactory



Cook's distance shows no unduly influential points (none with $D > 1$).

Observation with greatest influence on the fit (also with largest residual in this case) is that corresponding to 'Human'.